

ANNEXE : Formulaire de trigonométrie

Angles particuliers

x	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π
cos x	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1
sin x	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0
tan x	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	//	0

Premières formules

$$\tan a = \frac{\sin a}{\cos a}$$

$$\cot a = \frac{1}{\tan a} = \frac{\cos a}{\sin a}$$

$$\cos^2 a + \sin^2 a = 1$$

$$1 + \tan^2 a = \frac{1}{\cos^2 a}$$

$$1 + \cot^2 a = \frac{1}{\sin^2 a}$$

Dérivées

$$\cos' x = -\sin x$$

$$\sin' x = \cos x$$

$$\tan' x = 1 + \tan^2 x = \frac{1}{\cos^2 x}$$

$$\cot' x = -1 - \cot^2 x = -\frac{1}{\sin^2 x}$$

Parité et symétries

$$\cos(-a) = \cos a$$

$$\sin(-a) = -\sin a$$

$$\tan(-a) = -\tan a$$

$$\cos(\pi - a) = -\cos a$$

$$\sin(\pi - a) = \sin a$$

$$\tan(\pi - a) = -\tan a$$

$$\cos(\pi + a) = -\cos a$$

$$\sin(\pi + a) = -\sin a$$

$$\tan(\pi + a) = \tan a$$

$$\cos\left(\frac{\pi}{2} - a\right) = \sin a$$

$$\sin\left(\frac{\pi}{2} - a\right) = \cos a$$

$$\tan\left(\frac{\pi}{2} - a\right) = \cot a$$

$$\cos\left(\frac{\pi}{2} + a\right) = -\sin a$$

$$\sin\left(\frac{\pi}{2} + a\right) = \cos a$$

$$\tan\left(\frac{\pi}{2} + a\right) = -\cot a$$

Formules de développement

$$\cos(a + b) = \cos a \cos b - \sin a \sin b$$

$$\cos(a - b) = \cos a \cos b + \sin a \sin b$$

$$\sin(a + b) = \sin a \cos b + \sin b \cos a$$

$$\sin(a - b) = \sin a \cos b - \sin b \cos a$$

$$\tan(a + b) = \frac{\tan a + \tan b}{1 - \tan a \tan b}$$

$$\tan(a - b) = \frac{\tan a - \tan b}{1 + \tan a \tan b}$$

Formules de duplication :

$$\begin{aligned}\cos(2a) &= \cos^2 a - \sin^2 a = 1 - 2\sin^2 a = 2\cos^2 a - 1 \\ \sin(2a) &= 2\sin a \cos a \\ \tan(2a) &= \frac{2\tan a}{1 - \tan^2 a}\end{aligned}$$

Transformation de produits en sommes

$$\begin{aligned}\cos a \cos b &= \frac{1}{2}(\cos(a+b) + \cos(a-b)) \\ \sin a \sin b &= \frac{1}{2}(\cos(a-b) - \cos(a+b)) \\ \sin a \cos b &= \frac{1}{2}(\sin(a+b) + \sin(a-b))\end{aligned}$$

Linéarisation :

$$\cos^2 a = \frac{1 + \cos 2a}{2} \qquad \sin^2 a = \frac{1 - \cos 2a}{2}$$

Transformation de sommes en produits

$$\begin{aligned}\cos p + \cos q &= 2\cos \frac{p+q}{2} \cos \frac{p-q}{2} \\ \cos p - \cos q &= -2\sin \frac{p+q}{2} \sin \frac{p-q}{2} \\ \sin p + \sin q &= 2\sin \frac{p+q}{2} \cos \frac{p-q}{2} \\ \sin p - \sin q &= 2\cos \frac{p+q}{2} \sin \frac{p-q}{2}\end{aligned}$$

Arc moitié. Posons $t = \tan \frac{a}{2}$.

$$\cos a = \frac{1 - t^2}{1 + t^2} \qquad \sin a = \frac{2t}{1 + t^2} \qquad \tan a = \frac{2t}{1 - t^2}$$

Factorisation d'une combinaison linéaire de $\cos x$ et $\sin x$ (avec $(a, b) \neq (0, 0)$) :

$$a \cos x + b \sin x = \sqrt{a^2 + b^2} \cos(x - \theta) \quad \text{où} \quad \begin{cases} \cos \theta = \frac{a}{\sqrt{a^2 + b^2}} \\ \sin \theta = \frac{b}{\sqrt{a^2 + b^2}} \end{cases}$$

Équations trigonométriques

$$\begin{aligned}\sin x &= \sin a \iff x \equiv a [2\pi] \text{ ou } x \equiv \pi - a [2\pi] \\ \cos x &= \cos a \iff x \equiv a [2\pi] \text{ ou } x \equiv -a [2\pi] \\ \cos x &= \cos a \text{ et } \sin x = \sin a \iff x \equiv a [2\pi] \\ \tan x &= \tan a \iff x \equiv a [\pi]\end{aligned}$$